EXERCISE 7

Seven Linear Systems Solved by Elimination

Original statement · probls_ch4-i_2627_v2.pdf, p. 2

Elimination Method • Seven IVPs • Undetermined Coefficients

STATEMENT

Find the general solutions of the following differential equations by converting them into a single second-order equation. Also find the solution that satisfies the given initial conditions.

- (i) $\dot{x} = 4x - y, \quad \dot{y} = 2x + y + t^2, \quad x(0) = 0,\ y(0) = 1;$
- (ii) $\dot{x} = x - 4y + \cos 2t, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = 1;$
- (iii) $\dot{x} = 2x + 2y, \quad \dot{y} = 6x + 3y + e^t, \quad x(0) = 0,\ \dot{x}(0) = 1;$
- (iv) $\dot{x} = 5x - 4y + e^{3t}, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = -1;$
- (v) $\dot{x} = 2x + 5y, \quad \dot{y} = -2x + \cos 3t, \quad x(0) = 2,\ y(0) = -1;$
- (vi) $\dot{x} = x + y + e^{-t}, \quad \dot{y} = 4x - 2y + e^{2t}, \quad x(0) = 1,\ y(0) = -1;$
- (vii) $\dot{x} = 8x + 14y, \quad \dot{y} = 7x + y, \quad x(0) = y(0) = 1.$

PHASE 1: SETUP AND CLASSIFICATION

Each system is linear with constant coefficients, of the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1), homogeneous when $f = g = 0$. In every item $b \neq 0$, so the first equation can be solved for y. Hypotheses: coefficients and forcings continuous on $\mathbb{R}$, so the IVP has a unique solution, which exists on all of $\mathbb{R}$ (linear system with continuous coefficients; Thm 25.2 gives local existence and uniqueness and the continuation argument of Exercise 4 extends it). Item (iii) gives $x(0)$ and $\dot{x}(0)$; the value of $y(0)$ then follows from the first equation.

PHASE 2: METHOD: ELIMINATION (TOPIC 4 SEC. 2.1)

Solve the first equation for $y = (\dot{x} - ax - f)/b$ (26.2), differentiate, and substitute in the second equation. The result is one second-order equation for x (26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Its coefficients are the trace and determinant of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, so its characteristic polynomial is the one of the eigenvalue method. It is solved by the methods of Topic 3 (complementary function plus undetermined coefficients). Then y is recovered from (26.2) and the constants from the data.

METHOD OF THE CLASS NOTES [NOTES, P. 9]

Method 1 of the notes: for $\dot{x} = ax + by, \dot{y} = cx + dy$ with $b \neq 0$ (true in all seven items), $y = (\dot{x} - ax)/b$ and $\dot{y} = (\ddot{x} - a\dot{x})/b$; substituting and multiplying by b gives $\ddot{x} - (a+d)\dot{x} + (ad - bc)x = 0$, the homogeneous part of the equation above. The forcing terms are a complement (not in the class notes); the second-order equation is then solved as in Topic 3, $x = x_H + x_N$ with the educated guess.

Discrepancy with the notes: they box $\ddot{x} - (a+b)\dot{x} + (ad - bc)x = 0$, but the steps give $a + d$. Check with $x' = 2x - y, y' = -x + 3y$: elimination gives $\ddot{x} - 5\dot{x} + 5x = 0$, and $a + d = 5$ while $a + b = 1$.

PHASE 3: DERIVATION OF THE SEVEN SOLUTIONS

(i) $\dot{x} = 4x - y, \quad \dot{y} = 2x + y + t^2, \quad x(0) = 0,\ y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 4, \quad b = -1, \quad c = 2, \quad d = 1, \quad f(t) = 0, \quad g(t) = t^2$$

Elimination of y

Since $b = -1 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-1}\left(\dot{x} - (4)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-1}\left(\ddot{x} - (4)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 5$, $ad - bc = (4)(1) - (-1)(2) = 6$ and $F(t) := \dot{f} + bg - df = (0) + (-1)(t^2) - (1)(0) = -t^2$, so

$$\ddot{x} - 5\dot{x} + 6x = -t^2$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (4)(0) + (-1)(1) + (0) = -1$.

The homogeneous equation $\ddot{x} - 5\dot{x} + 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 5k + 6 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-5)^2 - 4(1)(6) = 1$.

Factorising, $P(k) = (k-3)\,(k-2)$, so the roots are

$$k = 2, \qquad k = 3$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{2t}, e^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{2t} + c_2 e^{3t}$$

The forcing $-t^2$ is a polynomial of degree 2; a polynomial is e^{0t} times a polynomial, so the exponent to test is $s = 0$: $P(0) = 6$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 + K_2 t + K_3 t^2$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_2 + 2K_3 t \\ x_p''(t) &= 2K_3 \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 5x_p' + 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 6K_3\, t^2 + (6K_2 - 10K_3)\, t + (6K_1 - 5K_2 + 2K_3)$$

This must equal the forcing $F(t) = -t^2$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} t^2: &\quad 6K_3 = -1 \\ t: &\quad 6K_2 - 10K_3 = 0 \\ 1: &\quad 6K_1 - 5K_2 + 2K_3 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} 6K_3 &= -1 \quad (1) \\ 6K_2 - 10K_3 &= 0 \quad (2) \\ 6K_1 - 5K_2 + 2K_3 &= 0 \quad (3) \end{aligned}$$

$6K_3 = -1$ contains one unknown only, so $K_3 = -\frac{1}{6}$.

Substitute in $6K_2 - 10K_3 = 0$: $6K_2 + \frac{5}{3} = 0$.

Substitute in $6K_1 - 5K_2 + 2K_3 = 0$: $6K_1 - 5K_2 - \frac{1}{3} = 0$.

$6K_2 + \frac{5}{3} = 0$ contains one unknown only, so $K_2 = -\frac{5}{18}$.

Substitute in $6K_1 - 5K_2 - \frac{1}{3} = 0$: $6K_1 + \frac{19}{18} = 0$.

$6K_1 + \frac{19}{18} = 0$ contains one unknown only, so $K_1 = -\frac{19}{108}$.

Hence $K_1 = -\frac{19}{108}, K_2 = -\frac{5}{18}, K_3 = -\frac{1}{6}$.

Therefore

$$x_p(t) = -\frac{t^2}{6} - \frac{5t}{18} - \frac{19}{108}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{2t} + c_2 e^{3t} - \frac{t^2}{6} - \frac{5t}{18} - \frac{19}{108}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 2c_1 e^{2t} + 3c_2 e^{3t} - \frac{t}{3} - \frac{5}{18}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{19}{108} = 0$$
$$x'(0) = 2c_1 + 3c_2 - \frac{5}{18} = -1$$

The linear system is

$$c_1 + c_2 - \frac{19}{108} = 0 \quad (1)$$
$$2c_1 + 3c_2 - \frac{5}{18} = -1 \quad (2)$$

Solve $c_1 + c_2 - \frac{19}{108} = 0$ for c_1: $c_1 = \frac{19}{108} - c_2$.

Substitute in $2c_1 + 3c_2 - \frac{5}{18} = -1$: $c_2 + \frac{2}{27} = -1$.

$c_2 + \frac{2}{27} = -1$ contains one unknown only, so $c_2 = -\frac{29}{27}$.

Back-substitution: $c_1 = \frac{19}{108} - c_2 = \frac{5}{4}$.

Hence $c_1 = \frac{5}{4}, c_2 = -\frac{29}{27}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{t^2}{6} - \frac{5t}{18} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{4} - \frac{19}{108}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18}$$

$$y(t) = \frac{1}{-1}\left(\dot{x} - (4)x - (0)\right) = -\frac{2t^2}{3} - \frac{7t}{9} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{2} - \frac{23}{54}$$

Check in the original system

$$\begin{aligned} \dot{x} &= -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18} \\ ax + by + f &= -\frac{t}{3} - \frac{29e^{3t}}{9} + \frac{5e^{2t}}{2} - \frac{5}{18} \\ \dot{y} &= -\frac{4t}{3} - \frac{29e^{3t}}{9} + 5e^{2t} - \frac{7}{9} \\ cx + dy + g &= -\frac{4t}{3} - \frac{29e^{3t}}{9} + 5e^{2t} - \frac{7}{9} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 0, y(0) = 1$ reproduce the initial data.

(ii) $\dot{x} = x - 4y + \cos 2t, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 1, \quad b = -4, \quad c = 1, \quad d = 1, \quad f(t) = \cos 2t, \quad g(t) = 0$$

Elimination of y

Since $b = -4 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-4}\left(\dot{x} - (1)x - (\cos 2t)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-4}\left(\ddot{x} - (1)\dot{x} - (-2\sin 2t)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 2$, $ad - bc = (1)(1) - (-4)(1) = 5$ and $F(t) := \dot{f} + bg - df = (-2\sin 2t) + (-4)(0) - (1)(\cos 2t) = -2\sin 2t - \cos 2t$, so

$$\ddot{x} - 2\dot{x} + 5x = -2\sin 2t - \cos 2t$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (1)(1) + (-4)(1) + (1) = -2$.

The homogeneous equation $\ddot{x} - 2\dot{x} + 5x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 2k + 5 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-2)^2 - 4(1)(5) = -16$.

Factorising, $P(k) = (k - 1 - 2i)\,(k - 1 + 2i)$, so the roots are

$$k = 1 - 2i, \qquad k = 1 + 2i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^t \cos 2t$, $e^t \sin 2t$, so the complementary function is

$$x_h(t) = c_1 e^t \cos 2t + c_2 e^t \sin 2t$$

The forcing $-2\sin 2t - \cos 2t$ is a sinusoidal term: it is the real or imaginary part of $e^{(2i)t}$, so we test $s = 2i$ in the characteristic polynomial: $P(2i) = 1 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos 2t + K_2 \sin 2t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -2K_1 \sin 2t + 2K_2 \cos 2t \\ x_p''(t) &= -4K_1 \cos 2t - 4K_2 \sin 2t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 2x_p' + 5x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (K_1 - 4K_2)\,\cos 2t + (4K_1 + K_2)\,\sin 2t$$

This must equal the forcing $F(t) = -2\sin 2t - \cos 2t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} \cos 2t: &\quad K_1 - 4K_2 = -1 \\ \sin 2t: &\quad 4K_1 + K_2 = -2 \end{aligned}$$

The linear system is

$$K_1 - 4K_2 = -1 \quad (1)$$
$$4K_1 + K_2 = -2 \quad (2)$$

Solve $K_1 - 4K_2 = -1$ for K_1: $K_1 = 4K_2 - 1$.

Substitute in $4K_1 + K_2 = -2$: $17K_2 - 4 = -2$.

$17K_2 - 4 = -2$ contains one unknown only, so $K_2 = \frac{2}{17}$.

Back-substitution: $K_1 = 4K_2 - 1 = -\frac{9}{17}$.

Hence $K_1 = -\frac{9}{17}, K_2 = \frac{2}{17}$.

Therefore

$$x_p(t) = \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos 2t + c_2 e^t \sin 2t + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -2c_1 e^t \sin 2t + c_1 e^t \cos 2t + c_2 e^t \sin 2t + 2c_2 e^t \cos 2t + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 - \frac{9}{17} = 1$$
$$x'(0) = c_1 + 2c_2 + \frac{4}{17} = -2$$

The linear system is

$$c_1 - \frac{9}{17} = 1 \quad (1)$$
$$c_1 + 2c_2 + \frac{4}{17} = -2 \quad (2)$$

$c_1 - \frac{9}{17} = 1$ contains one unknown only, so $c_1 = \frac{26}{17}$.

Substitute in $c_1 + 2c_2 + \frac{4}{17} = -2$: $2c_2 + \frac{30}{17} = -2$.

$2c_2 + \frac{30}{17} = -2$ contains one unknown only, so $c_2 = -\frac{32}{17}$.

Hence $c_1 = \frac{26}{17}$, $c_2 = -\frac{32}{17}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{32e^t \sin 2t}{17} + \frac{26e^t \cos 2t}{17} + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17}$$

$$y(t) = \frac{1}{-4}\left(\dot{x} - (1)x - (\cos 2t)\right) = \frac{13e^t \sin 2t}{17} + \frac{16e^t \cos 2t}{17} - \frac{4\sin 2t}{17} + \frac{\cos 2t}{17}$$

Check in the original system

$$\begin{aligned} \dot{x} &= -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17} \\ ax + by + f &= -\frac{84e^t \sin 2t}{17} - \frac{38e^t \cos 2t}{17} + \frac{18\sin 2t}{17} + \frac{4\cos 2t}{17} \\ \dot{y} &= -\frac{19e^t \sin 2t}{17} + \frac{42e^t \cos 2t}{17} - \frac{2\sin 2t}{17} - \frac{8\cos 2t}{17} \\ cx + dy + g &= -\frac{19e^t \sin 2t}{17} + \frac{42e^t \cos 2t}{17} - \frac{2\sin 2t}{17} - \frac{8\cos 2t}{17} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = 1$ reproduce the initial data.

(iii) $\dot{x} = 2x + 2y, \quad \dot{y} = 6x + 3y + e^t, \quad x(0) = 0,\ \dot{x}(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 2, \quad b = 2, \quad c = 6, \quad d = 3, \quad f(t) = 0, \quad g(t) = e^t$$

Elimination of y

Since $b = 2 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - a x - f}{b} = \frac{1}{2}\left(\dot{x} - (2)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{2}\left(\ddot{x} - (2)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - a x - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 5$, $ad - bc = (2)(3) - (2)(6) = -6$ and $F(t) := \dot{f} + bg - df = (0) + (2)(e^{t}) - (3)(0) = 2e^{t}$, so

$$\ddot{x} - 5\dot{x} - 6x = 2e^{t}$$

Initial data. Here $x(0)$ and $\dot{x}(0)$ are given; the value of $y(0)$ follows from the first equation at $t = 0$: $y(0) = \dfrac{\dot{x}(0) - a\,x(0) - f(0)}{b} = \dfrac{1 - (2)(0) - (0)}{2} = \tfrac{1}{2}$.

The homogeneous equation $\ddot{x} - 5\dot{x} - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^{2} - 5k - 6 = 0$$

Its discriminant is $\Delta = b^{2} - 4ac = (-5)^{2} - 4(1)(-6) = 49$.

Factorising, $P(k) = (k - 6)\,(k + 1)$, so the roots are

$$k = -1, \qquad k = 6$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-t}, e^{6t}, so the complementary function is

$$x_h(t) = c_1 e^{-t} + c_2 e^{6t}$$

The forcing $2e^{t}$ is an exponential with exponent $s = 1$; we test it in the characteristic polynomial: $P(1) = -10$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= K_1 e^{t} \\ x_p''(t) &= K_1 e^{t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 5x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -10K_1\, e^{t}$$

This must equal the forcing $F(t) = 2e^{t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{t}: \quad -10K_1 = 2$$

The linear system is

$$-10K_1 = 2 \quad (1)$$

$-10K_1 = 2$ contains one unknown only, so $K_1 = -\frac{1}{5}$.

Hence $K_1 = -\frac{1}{5}$.

Therefore

$$x_p(t) = -\frac{e^t}{5}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-t} + c_2 e^{6t} - \frac{e^t}{5}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -c_1 e^{-t} + 6c_2 e^{6t} - \frac{e^t}{5}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{1}{5} = 0$$
$$x'(0) = -c_1 + 6c_2 - \frac{1}{5} = 1$$

The linear system is

$$c_1 + c_2 - \frac{1}{5} = 0 \quad (1)$$
$$-c_1 + 6c_2 - \frac{1}{5} = 1 \quad (2)$$

Solve $c_1 + c_2 - \frac{1}{5} = 0$ for c_1: $c_1 = \frac{1}{5} - c_2$.

Substitute in $-c_1 + 6c_2 - \frac{1}{5} = 1$: $7c_2 - \frac{2}{5} = 1$.

$7c_2 - \frac{2}{5} = 1$ contains one unknown only, so $c_2 = \frac{1}{5}$.

Back-substitution: $c_1 = \frac{1}{5} - c_2 = 0$.

Hence $c_1 = 0$, $c_2 = \frac{1}{5}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{e^{6t}}{5} - \frac{e^{t}}{5}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = \frac{6e^{6t}}{5} - \frac{e^{t}}{5}$$

$$y(t) = \frac{1}{2}\left(\dot{x} - (2)x - (0)\right) = \frac{2e^{6t}}{5} + \frac{e^{t}}{10}$$

Check in the original system

$$\begin{aligned} \dot{x} &= \frac{6e^{6t}}{5} - \frac{e^{t}}{5} \\ ax + by + f &= \frac{6e^{6t}}{5} - \frac{e^{t}}{5} \\ \dot{y} &= \frac{12e^{6t}}{5} + \frac{e^{t}}{10} \\ cx + dy + g &= \frac{12e^{6t}}{5} + \frac{e^{t}}{10} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 0$, $y(0) = \frac{1}{2}$ reproduce the initial data.

(iv) $\dot{x} = 5x - 4y + e^{3t}, \quad \dot{y} = x + y, \quad x(0) = 1,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t)$, $\dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 5, \quad b = -4, \quad c = 1, \quad d = 1, \quad f(t) = e^{3t}, \quad g(t) = 0$$

Elimination of y

Since $b = -4 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{-4}\left(\dot{x} - (5)x - (e^{3t})\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{-4}\left(\ddot{x} - (5)\dot{x} - (3e^{3t})\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\,(\dot{x} - ax - f) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a + d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 6$, $ad - bc = (5)(1) - (-4)(1) = 9$ and $F(t) := \dot{f} + bg - df = (3e^{3t}) + (-4)(0) - (1)(e^{3t}) = 2e^{3t}$, so

$$\ddot{x} - 6\dot{x} + 9x = 2e^{3t}$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (5)(1) + (-4)(-1) + (1) = 10$.

The homogeneous equation $\ddot{x} - 6\dot{x} + 9x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 6k + 9 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-6)^2 - 4(1)(9) = 0$.

Factorising, $P(k) = (k - 3)^2$, so the roots are

$$k = 3 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{3t}, te^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{3t} + c_2 t e^{3t}$$

The forcing $2e^{3t}$ is an exponential with exponent $s = 3$; we test it in the characteristic polynomial: $P(3) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 2, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^2.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t^2 e^{3t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= 3K_1 t^2 e^{3t} + 2K_1 t e^{3t} \\ x_p''(t) &= 9K_1 t^2 e^{3t} + 12K_1 t e^{3t} + 2K_1 e^{3t} \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 6x_p' + 9x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 2K_1\, e^{3t}$$

This must equal the forcing $F(t) = 2e^{3t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{3t}: \quad 2K_1 = 2$$

The linear system is

$$2K_1 = 2 \quad (1)$$

$2K_1 = 2$ contains one unknown only, so $K_1 = 1$.

Hence $K_1 = 1$.

Therefore

$$x_p(t) = t^2 e^{3t}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{3t} + c_2 t e^{3t} + t^2 e^{3t}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = 3c_1 e^{3t} + 3c_2 t e^{3t} + c_2 e^{3t} + 3t^2 e^{3t} + 2te^{3t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 = 1$$
$$x'(0) = 3c_1 + c_2 = 10$$

The linear system is

$$c_1 = 1 \quad (1)$$
$$3c_1 + c_2 = 10 \quad (2)$$

$c_1 = 1$ contains one unknown only, so $c_1 = 1$.

Substitute in $3c_1 + c_2 = 10$: $c_2 + 3 = 10$.

$c_2 + 3 = 10$ contains one unknown only, so $c_2 = 7$.

Hence $c_1 = 1, c_2 = 7$.

Substituting the constants into the general solution,

$$\boxed{x(t) = t^2 e^{3t} + 7te^{3t} + e^{3t}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = 3t^2 e^{3t} + 23te^{3t} + 10e^{3t}$$

$$y(t) = \frac{1}{-4}\left(\dot{x} - (5)x - (e^{3t})\right) = \frac{t^2 e^{3t}}{2} + 3te^{3t} - e^{3t}$$

Check in the original system

$$\begin{aligned} \dot{x} &= 3t^2 e^{3t} + 23te^{3t} + 10e^{3t} \\ ax + by + f &= 3t^2 e^{3t} + 23te^{3t} + 10e^{3t} \\ \dot{y} &= \frac{3t^2 e^{3t}}{2} + 10te^{3t} \\ cx + dy + g &= \frac{3t^2 e^{3t}}{2} + 10te^{3t} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = -1$ reproduce the initial data.

(v) $\dot{x} = 2x + 5y, \quad \dot{y} = -2x + \cos 3t, \quad x(0) = 2,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 2, \quad b = 5, \quad c = -2, \quad d = 0, \quad f(t) = 0, \quad g(t) = \cos 3t$$

Elimination of y

Since $b = 5 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{5}\left(\dot{x} - (2)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{5}\left(\ddot{x} - (2)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 2$, $ad - bc = (2)(0) - (5)(-2) = 10$ and $F(t) := \dot{f} + bg - df = (0) + (5)(\cos 3t) - (0)(0) = 5\cos 3t$, so

$$\ddot{x} - 2\dot{x} + 10x = 5\cos 3t$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (2)(2) + (5)(-1) + (0) = -1$.

The homogeneous equation $\ddot{x} - 2\dot{x} + 10x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 2k + 10 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-2)^2 - 4(1)(10) = -36$.

Factorising, $P(k) = (k - 1 - 3i)\,(k - 1 + 3i)$, so the roots are

$$k = 1 - 3i, \qquad k = 1 + 3i$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{t}\cos 3t$, $e^{t}\sin 3t$, so the complementary function is

$$x_h(t) = c_1 e^{t}\cos 3t + c_2 e^{t}\sin 3t$$

The forcing $5\cos 3t$ is a sinusoidal term: it is the real or imaginary part of $e^{(3i)t}$, so we test $s = 3i$ in the characteristic polynomial: $P(3i) = 1 - 6i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos 3t + K_2 \sin 3t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -3K_1 \sin 3t + 3K_2 \cos 3t \\ x_p''(t) &= -9K_1 \cos 3t - 9K_2 \sin 3t \end{aligned}$$

Substitute into $L[x_p] = x_p'' - 2x_p' + 10x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (K_1 - 6K_2)\, \cos 3t + (6K_1 + K_2)\, \sin 3t$$

This must equal the forcing $F(t) = 5\cos 3t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$\begin{aligned} \cos 3t: &\quad K_1 - 6K_2 = 5 \\ \sin 3t: &\quad 6K_1 + K_2 = 0 \end{aligned}$$

The linear system is

$$\begin{aligned} K_1 - 6K_2 &= 5 \quad (1) \\ 6K_1 + K_2 &= 0 \quad (2) \end{aligned}$$

Solve $K_1 - 6K_2 = 5$ for K_1: $K_1 = 6K_2 + 5$.

Substitute in $6K_1 + K_2 = 0$: $37K_2 + 30 = 0$.

$37K_2 + 30 = 0$ contains one unknown only, so $K_2 = -\frac{30}{37}$.

Back-substitution: $K_1 = 6K_2 + 5 = \frac{5}{37}$.

Hence $K_1 = \frac{5}{37}, K_2 = -\frac{30}{37}$.

Therefore

$$x_p(t) = -\frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos 3t + c_2 e^t \sin 3t - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^t \sin 3t + c_1 e^t \cos 3t + c_2 e^t \sin 3t + 3c_2 e^t \cos 3t - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 + \frac{5}{37} = 2 \\ x'(0) &= c_1 + 3c_2 - \frac{90}{37} = -1 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 + \frac{5}{37} &= 2 \quad (1) \\ c_1 + 3c_2 - \frac{90}{37} &= -1 \quad (2) \end{aligned}$$

$c_1 + \frac{5}{37} = 2$ contains one unknown only, so $c_1 = \frac{69}{37}$.

Substitute in $c_1 + 3c_2 - \frac{90}{37} = -1$: $3c_2 - \frac{21}{37} = -1$.

$3c_2 - \frac{21}{37} = -1$ contains one unknown only, so $c_2 = -\frac{16}{111}$.

Hence $c_1 = \frac{69}{37}$, $c_2 = -\frac{16}{111}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = -\frac{16e^t \sin 3t}{111} + \frac{69e^t \cos 3t}{37} - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37}$$

$$y(t) = \frac{1}{5}\left(\dot{x} - (2)x - (0)\right) = -\frac{121e^{t}\sin 3t}{111} - \frac{17e^{t}\cos 3t}{37} + \frac{9\sin 3t}{37} - \frac{20\cos 3t}{37}$$

Check in the original system

$$\begin{aligned}
\dot{x} &= -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37} \\
ax + by + f &= -\frac{637e^{t}\sin 3t}{111} + \frac{53e^{t}\cos 3t}{37} - \frac{15\sin 3t}{37} - \frac{90\cos 3t}{37} \\
\dot{y} &= \frac{32e^{t}\sin 3t}{111} - \frac{138e^{t}\cos 3t}{37} + \frac{60\sin 3t}{37} + \frac{27\cos 3t}{37} \\
cx + dy + g &= \frac{32e^{t}\sin 3t}{111} - \frac{138e^{t}\cos 3t}{37} + \frac{60\sin 3t}{37} + \frac{27\cos 3t}{37}
\end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 2, y(0) = -1$ reproduce the initial data.

(vi) $\dot{x} = x + y + e^{-t}, \quad \dot{y} = 4x - 2y + e^{2t}, \quad x(0) = 1,\ y(0) = -1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 1, \quad b = 1, \quad c = 4, \quad d = -2, \quad f(t) = e^{-t}, \quad g(t) = e^{2t}$$

Elimination of y

Since $b = 1 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{1}\left(\dot{x} - (1)x - (e^{-t})\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{1}\left(\ddot{x} - (1)\dot{x} - (-e^{-t})\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = -1$, $ad - bc = (1)(-2) - (1)(4) = -6$ and $F(t) := \dot{f} + bg - df = (-e^{-t}) + (1)(e^{2t}) - (-2)(e^{-t}) = e^{2t} + e^{-t}$, so

$$\ddot{x} + \dot{x} - 6x = e^{2t} + e^{-t}$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (1)(1) + (1)(-1) + (1) = 1$.

The homogeneous equation $\ddot{x} + \dot{x} - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 + k - 6 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (1)^2 - 4(1)(-6) = 25$.

Factorising, $P(k) = (k - 2)\,(k + 3)$, so the roots are

$$k = -3, \qquad k = 2$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-3t}, e^{2t}, so the complementary function is

$$x_h(t) = c_1 e^{-3t} + c_2 e^{2t}$$

The forcing is a sum of terms. By linearity (Robinson Eq. 14.7) a particular integral is the sum of particular integrals of the separate terms, each found in turn.

Particular integral for the term e^{-t}

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -6$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = -K_1 e^{-t}$$
$$x_p''(t) = K_1 e^{-t}$$

Substitute into $L[x_p] = x_p'' + x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -6K_1\, e^{-t}$$

This must equal the forcing $F(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{-t}: \quad -6K_1 = 1$$

The linear system is

$$-6K_1 = 1 \quad (1)$$

$-6K_1 = 1$ contains one unknown only, so $K_1 = -\frac{1}{6}$.

Hence $K_1 = -\frac{1}{6}$.

Therefore

$$x_p(t) = -\frac{e^{-t}}{6}$$

Particular integral for the term e^{2t}

The forcing e^{2t} is an exponential with exponent $s = 2$; we test it in the characteristic polynomial: $P(2) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by t^1.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 t e^{2t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$x_p'(t) = 2K_1 t e^{2t} + K_1 e^{2t}$$
$$x_p''(t) = 4K_1 t e^{2t} + 4K_1 e^{2t}$$

Substitute into $L[x_p] = x_p'' + x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = 5K_1\, e^{2t}$$

This must equal the forcing $F(t) = e^{2t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in F:

$$e^{2t}: \quad 5K_1 = 1$$

The linear system is

$$5K_1 = 1 \quad (1)$$

$5K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{5}$.

Hence $K_1 = \frac{1}{5}$.

Therefore

$$x_p(t) = \frac{te^{2t}}{5}$$

Adding the particular integrals,

$$x_p(t) = \frac{te^{2t}}{5} - \frac{e^{-t}}{6}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{-3t} + c_2 e^{2t} + \frac{te^{2t}}{5} - \frac{e^{-t}}{6}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -3c_1 e^{-3t} + 2c_2 e^{2t} + \frac{2te^{2t}}{5} + \frac{e^{2t}}{5} + \frac{e^{-t}}{6}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$x(0) = c_1 + c_2 - \frac{1}{6} = 1$$
$$x'(0) = -3c_1 + 2c_2 + \frac{11}{30} = 1$$

The linear system is

$$c_1 + c_2 - \frac{1}{6} = 1 \quad (1)$$
$$-3c_1 + 2c_2 + \frac{11}{30} = 1 \quad (2)$$

Solve $c_1 + c_2 - \frac{1}{6} = 1$ for c_1: $c_1 = \frac{7}{6} - c_2$.

Substitute in $-3c_1 + 2c_2 + \frac{11}{30} = 1$: $5c_2 - \frac{47}{15} = 1$.

$5c_2 - \frac{47}{15} = 1$ contains one unknown only, so $c_2 = \frac{62}{75}$.

Back-substitution: $c_1 = \frac{7}{6} - c_2 = \frac{17}{50}$.

Hence $c_1 = \frac{17}{50}, c_2 = \frac{62}{75}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{te^{2t}}{5} + \frac{62e^{2t}}{75} - \frac{e^{-t}}{6} + \frac{17e^{-3t}}{50}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50}$$

$$y(t) = \frac{1}{1}\left(\dot{x} - (1)x - (e^{-t})\right) = \frac{te^{2t}}{5} + \frac{77e^{2t}}{75} - \frac{2e^{-t}}{3} - \frac{34e^{-3t}}{25}$$

Check in the original system

$$\begin{aligned} \dot{x} &= \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50} \\ ax + by + f &= \frac{2te^{2t}}{5} + \frac{139e^{2t}}{75} + \frac{e^{-t}}{6} - \frac{51e^{-3t}}{50} \\ \dot{y} &= \frac{2te^{2t}}{5} + \frac{169e^{2t}}{75} + \frac{2e^{-t}}{3} + \frac{102e^{-3t}}{25} \\ cx + dy + g &= \frac{2te^{2t}}{5} + \frac{169e^{2t}}{75} + \frac{2e^{-t}}{3} + \frac{102e^{-3t}}{25} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = -1$ reproduce the initial data.

(vii) $\dot{x} = 8x + 14y, \quad \dot{y} = 7x + y, \quad x(0) = y(0) = 1$

The system is linear with constant coefficients, in the form $\dot{x} = ax + by + f(t), \dot{y} = cx + dy + g(t)$ (Robinson Eq. 26.1) with

$$a = 8, \quad b = 14, \quad c = 7, \quad d = 1, \quad f(t) = 0, \quad g(t) = 0$$

Elimination of y

Since $b = 14 \neq 0$, the first equation can be solved for y (Eq. 26.2):

$$y = \frac{\dot{x} - ax - f}{b} = \frac{1}{14}\left(\dot{x} - (8)x - (0)\right)$$

Differentiate with respect to t (everything is differentiable):

$$\dot{y} = \frac{\ddot{x} - a\dot{x} - \dot{f}}{b} = \frac{1}{14}\left(\ddot{x} - (8)\dot{x} - (0)\right)$$

Substitute both expressions into the second equation, $\dot{y} = cx + dy + g$, and multiply by b:

$$\ddot{x} - a\dot{x} - \dot{f} = bc\,x + d\left(\dot{x} - ax - f\right) + bg$$

Collect the terms in $\ddot{x}$, $\dot{x}$ and x (Eq. 26.3):

$$\ddot{x} - (a+d)\dot{x} + (ad - bc)x = \dot{f} + bg - df$$

Numerically, $a + d = 9$, $ad - bc = (8)(1) - (14)(7) = -90$ and $F(t) := \dot{f} + bg - df = (0) + (14)(0) - (1)(0) = 0$, so

$$\ddot{x} - 9\dot{x} - 90x = 0$$

Initial data. The condition on y is converted into one on $\dot{x}$ with the first equation at $t = 0$: $\dot{x}(0) = a\,x(0) + b\,y(0) + f(0) = (8)(1) + (14)(1) + (0) = 22$.

The homogeneous equation $\ddot{x} - 9\dot{x} - 90x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^2 - 9k - 90 = 0$$

Its discriminant is $\Delta = b^2 - 4ac = (-9)^2 - 4(1)(-90) = 441$.

Factorising, $P(k) = (k - 15)\,(k + 6)$, so the roots are

$$k = -6, \qquad k = 15$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-6t}, e^{15t}, so the complementary function is

$$x_h(t) = c_1 e^{-6t} + c_2 e^{15t}$$

The equation is homogeneous, so there is no particular integral and the general solution is the complementary function:

$$x(t) = x_h(t) = c_1 e^{-6t} + c_2 e^{15t}$$

Initial conditions. Differentiate the general solution (product and chain rules):

$$x'(t) = -6c_1 e^{-6t} + 15c_2 e^{15t}$$

Evaluate at $t = 0$, using $e^0 = 1$, $\cos 0 = 1$ and $\sin 0 = 0$:

$$\begin{aligned} x(0) &= c_1 + c_2 = 1 \\ x'(0) &= -6c_1 + 15c_2 = 22 \end{aligned}$$

The linear system is

$$\begin{aligned} c_1 + c_2 &= 1 \quad (1) \\ -6c_1 + 15c_2 &= 22 \quad (2) \end{aligned}$$

Solve $c_1 + c_2 = 1$ for c_1: $c_1 = 1 - c_2$.

Substitute in $-6c_1 + 15c_2 = 22$: $21c_2 - 6 = 22$.

$21c_2 - 6 = 22$ contains one unknown only, so $c_2 = \frac{4}{3}$.

Back-substitution: $c_1 = 1 - c_2 = -\frac{1}{3}$.

Hence $c_1 = -\frac{1}{3}, c_2 = \frac{4}{3}$.

Substituting the constants into the general solution,

$$\boxed{x(t) = \frac{4e^{15t}}{3} - \frac{e^{-6t}}{3}}$$

Recovering y

From Eq. 26.2, $y = (\dot{x} - ax - f)/b$. With the solution found,

$$\dot{x}(t) = 20e^{15t} + 2e^{-6t}$$

$$y(t) = \frac{1}{14}\left(\dot{x} - (8)x - (0)\right) = \frac{2e^{15t}}{3} + \frac{e^{-6t}}{3}$$

Check in the original system

$$\begin{aligned} \dot{x} &= 20e^{15t} + 2e^{-6t} \\ ax + by + f &= 20e^{15t} + 2e^{-6t} \\ \dot{y} &= 10e^{15t} - 2e^{-6t} \\ cx + dy + g &= 10e^{15t} - 2e^{-6t} \end{aligned}$$

Both pairs of expressions coincide, and $x(0) = 1, y(0) = 1$ reproduce the initial data.

PHASE 4: INTERPRETATION AND CHECKS

Item	$x(t)$	$y(t)$
(i)	$-\frac{t^2}{6} - \frac{5t}{18} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{4} - \frac{19}{108}$	$-\frac{2t^2}{3} - \frac{7t}{9} - \frac{29e^{3t}}{27} + \frac{5e^{2t}}{2} - \frac{23}{54}$
(ii)	$-\frac{32e^t\sin 2t}{17} + \frac{26e^t\cos 2t}{17} + \frac{2\sin 2t}{17} - \frac{9\cos 2t}{17}$	$\frac{13e^t\sin 2t}{17} + \frac{16e^t\cos 2t}{17} - \frac{4\sin 2t}{17} + \frac{\cos 2t}{17}$
(iii)	$\frac{e^{6t}}{5} - \frac{e^t}{5}$	$\frac{2e^{6t}}{5} + \frac{e^t}{10}$
(iv)	$t^2e^{3t} + 7te^{3t} + e^{3t}$	$\frac{t^2e^{3t}}{2} + 3te^{3t} - e^{3t}$
(v)	$-\frac{16e^t\sin 3t}{111} + \frac{69e^t\cos 3t}{37} - \frac{30\sin 3t}{37} + \frac{5\cos 3t}{37}$	$-\frac{121e^t\sin 3t}{111} - \frac{17e^t\cos 3t}{37} + \frac{9\sin 3t}{37} - \frac{20\cos 3t}{37}$
(vi)	$\frac{te^{2t}}{5} + \frac{62e^{2t}}{75} - \frac{e^{-t}}{6} + \frac{17e^{-3t}}{50}$	$\frac{te^{2t}}{5} + \frac{77e^{2t}}{75} - \frac{2e^{-t}}{3} - \frac{34e^{-3t}}{25}$
(vii)	$\frac{4e^{15t}}{3} - \frac{e^{-6t}}{3}$	$\frac{2e^{15t}}{3} + \frac{e^{-6t}}{3}$

- Every pair (x, y) was substituted in **both** original equations and in the initial data (shown in each item and re-checked symbolically in `scripts/verify_topic3_topic4.py`).
- Characteristic exponents seen: (i) $2, 3$ (real); (ii) $1 \pm 2i$ (spiral) with the forcing $\cos 2t$ at a different frequency; (iii) $6, -1$ (saddle; the forcing e^t has exponent 1, which is **not** a root, so no extra factor t); (iv) 3 double (improper-node type), and the forcing e^{3t} resonates with the double root, giving t^2e^{3t}; (v) $1 \pm 3i$; (vi) $2, -3$ with the forcing exponent 2 resonant (term te^{2t}) and -1 not; (vii) $15, -6$ (homogeneous saddle).

- Cross-check of the homogeneous parts with Topic 4: (iv) and (v) have the same matrices as Examples 8.2 and 7.1; (vii) is the matrix of Exercise 9 (i).
- Dimensions: with t in seconds the matrix entries have units s^{-1} and the forcings those of $\dot{x}$; the elimination keeps this consistent ($a+d$ in s^{-1}, $ad-bc$ in s^{-2}).